



Hematology disorder

- Disseminated Intravascular Coagulation (DIC)

Prepared by

Dr / Ghada Elsayed

Disseminated Intravascular Coagulation (DIC) .

Outline

- Definition of DIC
- Factors Accelerating DIC
- Causes DIC
- Pathophysiology DIC
- Symptoms of DIC
- Diagnosis DIC
- Treatment DIC
- Nursing Care Plans & Management



Pre test

Mention
causes of DIC

Mention signs
and symptoms
of DIC

Disseminated intravascular coagulation

Introduction

is a life-threatening condition that requires immediate medical attention.

The formation of micro clots affects tissue perfusion in the major organs, causing hypoxia, ischemia, and tissue damage.

Coagulation occurs in two different pathways: intrinsic and extrinsic.

**These pathways
for the formation
of fibrin clots
and blood clotting.**

In the intrinsic pathway, endothelial cell damage commonly occurs because of sepsis or infection.

The extrinsic pathway is initiated by tissue injuries such as from malignancy, trauma, or obstetrical complications.

DIC may present as an acute or chronic condition.

Definition

- Disseminated intravascular coagulation is a condition in which blood clots form throughout the body, blocking small blood vessels.

Factors Accelerating DIC

☐ Shock

☐ Acidosis

☐ Hypoxemia

☐ Stasis

☐ Dehydration

☐ Hyperthermia



- ❑ Chronic renal insufficiency

- ❑ Chronic hepatic insufficiency

- ❑ Malnutrition

- ❑ Impaired anti-coagulation activity

- ❑ Impaired fibrinolytic activity

- ❑ Phagocytic dysfunction

Causes DIC

Cancer (particularly acute promyelocytic leukemia)

Complications of pregnancy: abruptio placentae, preeclampsia or eclampsia, postpartum hemorrhage

Massive tissue injury: trauma, burn, hyperthermia, rhabdomyolysis, surgery.

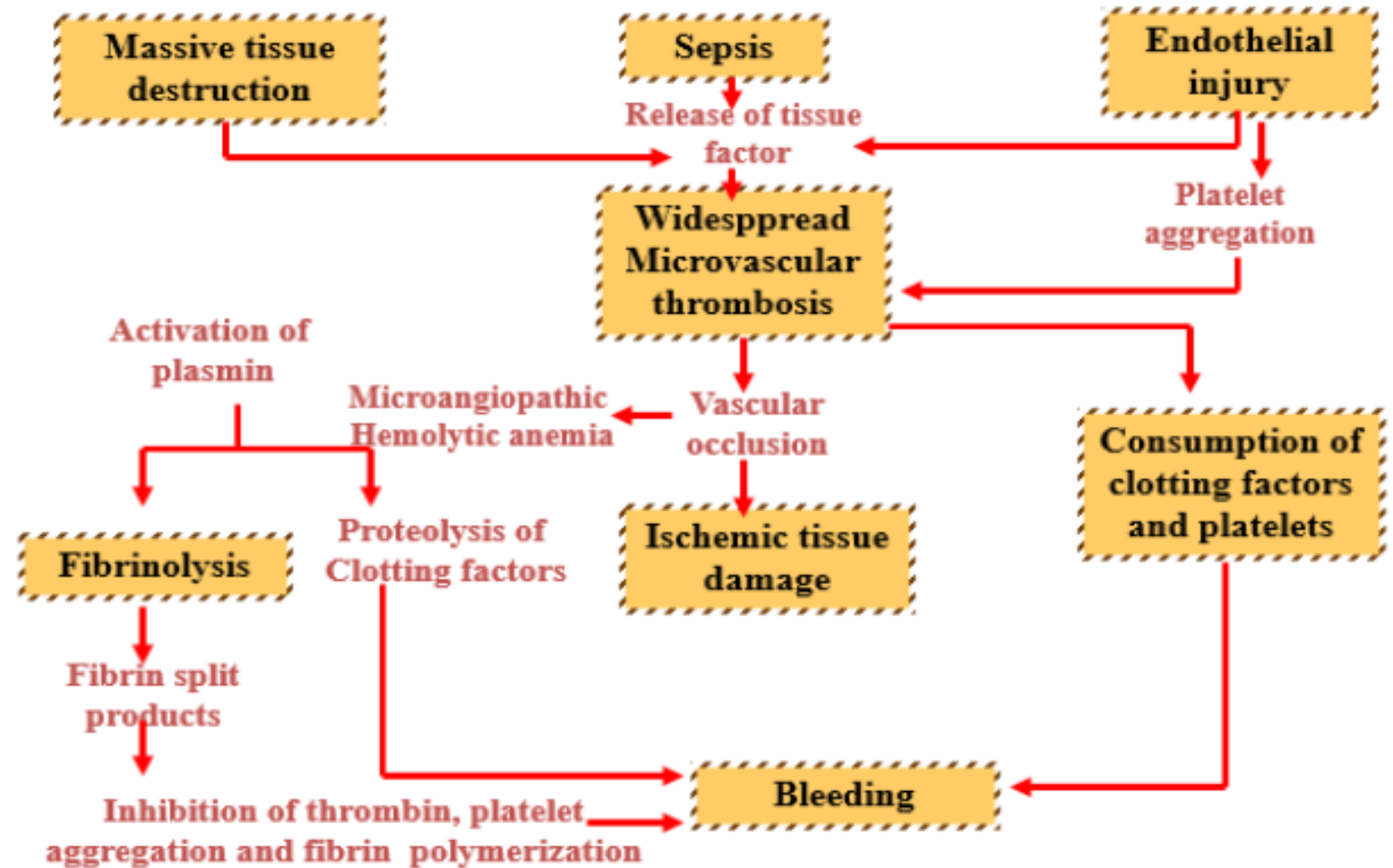
Infection: bacterial (Gram-negative or Gram-positive), viral, fungal, or protozoan

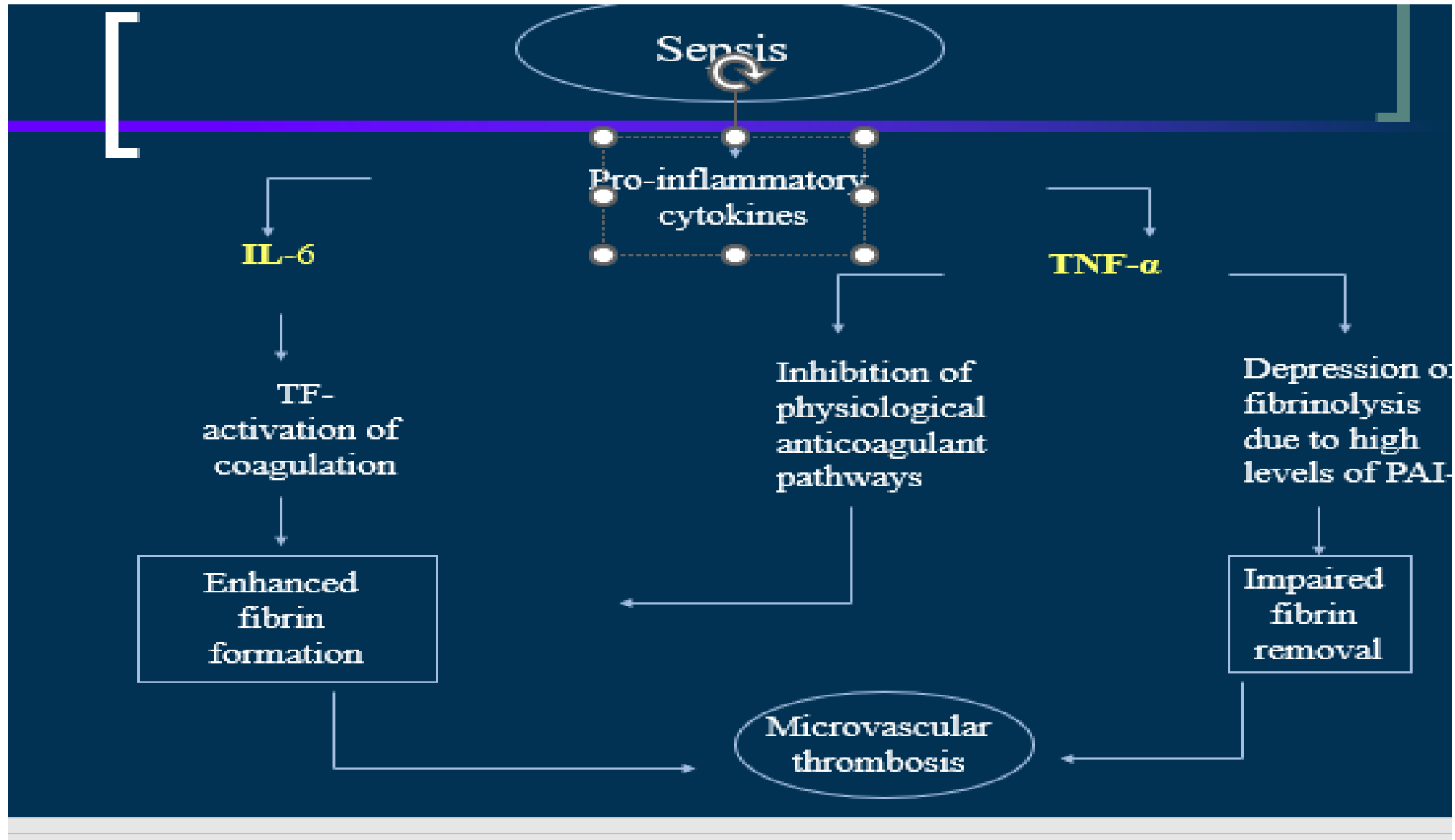
Transfusion reaction: ABO incompatibility

- 
- **Aortic aneurysm**
 - Liver disease , HELLP syndrome, thrombotic thrombocytopenic purpura, hemolytic uremic syndrome, and malignant hypertension may mimic DIC but originate via other pathways.
 - It is not the same as conditions where the blood vessels leak, such as capillary leak syndrome

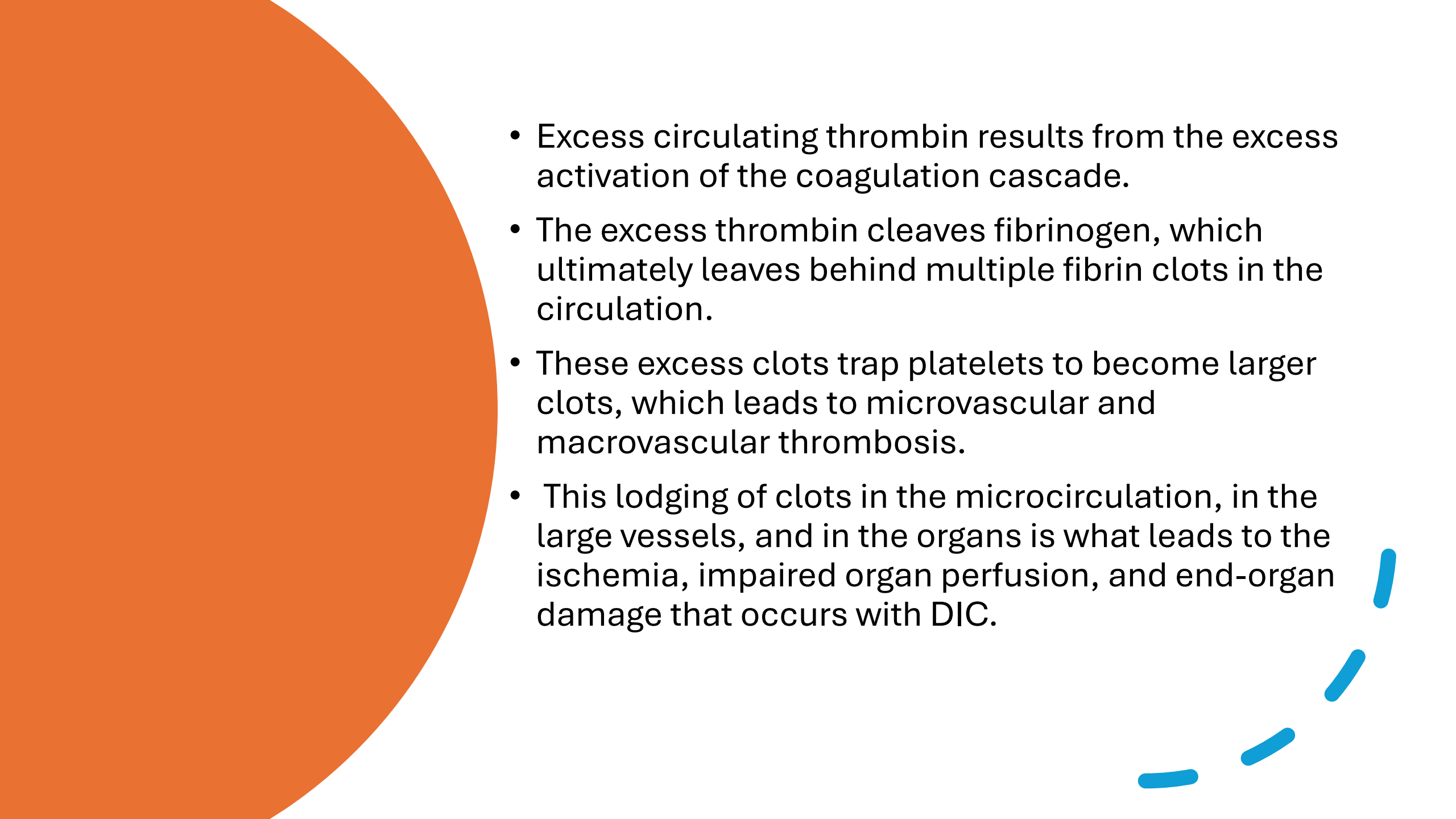
Pathophysiology

Pathophysiology of disseminated intravascular coagulation





-
- The fibrinolytic system then functions to break down fibrinogen and fibrin.
 - Activation of the fibrinolytic system generates plasmin (in the presence of thrombin), which is responsible for the lysis of fibrin clots.
 - The breakdown of fibrinogen and fibrin results in polypeptides called fibrin degradation products (FDPs) or fibrin split products (FSPs).
 - In DIC, the processes of coagulation and fibrinolysis are dysregulated, and the result is widespread clotting with resultant bleeding.

- 
- Excess circulating thrombin results from the excess activation of the coagulation cascade.
 - The excess thrombin cleaves fibrinogen, which ultimately leaves behind multiple fibrin clots in the circulation.
 - These excess clots trap platelets to become larger clots, which leads to microvascular and macrovascular thrombosis.
 - This lodging of clots in the microcirculation, in the large vessels, and in the organs is what leads to the ischemia, impaired organ perfusion, and end-organ damage that occurs with DIC.

Symptoms of DIC

Bleeding, from many sites in the body

Blood clots

Bruising

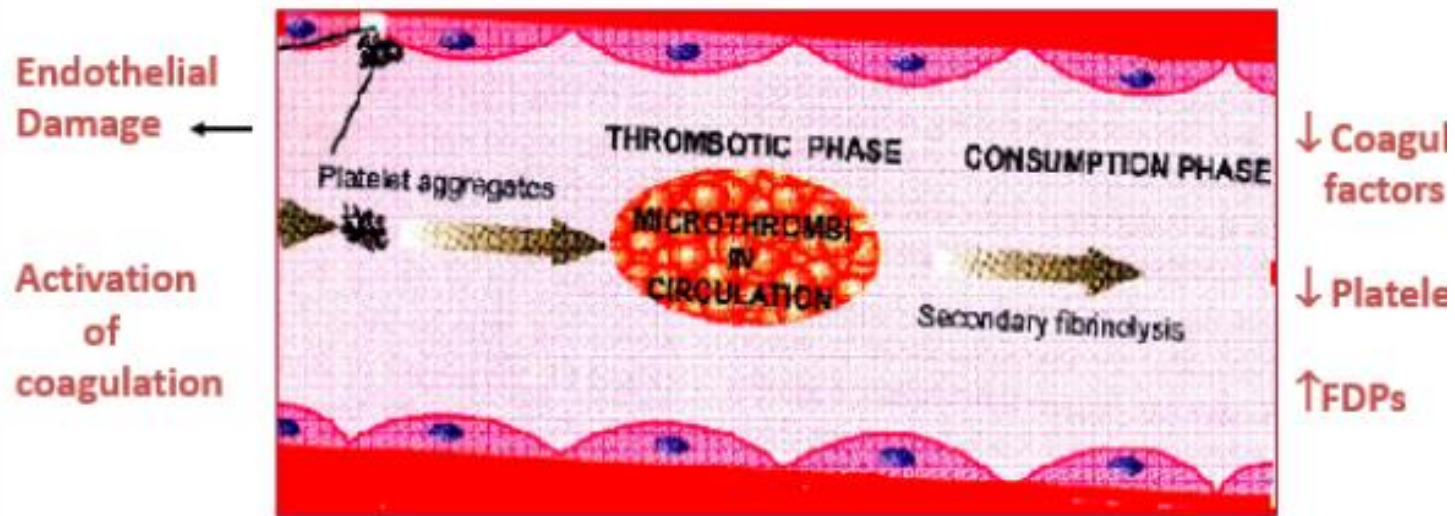
Drop in blood pressure

Shortness of breath

Confusion, memory loss or change of behavior

Fever

Disseminated Intravascular Coagulation (DIC)



• Diagnosis and diagnostic test

- Prolongation of the prothrombin time (PT) and the activated partial thromboplastin time (aPTT) reflect the underlying consumption and impaired synthesis of the coagulation cascade.
- **Fibrinogen level** was initially thought to be useful in the diagnosis of DIC but because it is an acute phase reactant, it will be **elevated** due to the underlying **inflammatory condition**.
- Therefore, a normal (or even elevated) level can occur in over 57% of cases
- . **A low level**, however, is more consistent with the consumptive process **of DIC**.



- A rapidly declining platelet count

High levels of fibrin degradation products.

- including D-dimer, are found owing to the intense fibrinolytic activity stimulated by the presence of fibrin in the circulation.
- The peripheral blood smear may show fragmented red blood cells (known as schistocytes) due to shear stress from thrombin. However, this finding is neither sensitive nor specific for DIC



A diagnostic algorithm has been proposed by the International Society of Thrombosis and Haemostasias.

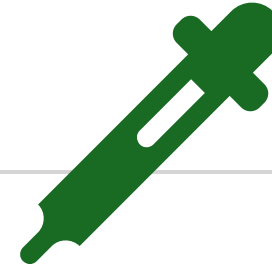
This algorithm appears to be 91% sensitive and 97% specific for the diagnosis of overt DIC.



A score of 5 or higher is compatible with DIC and it is recommended that the score is repeated daily, while a score below 5 is suggestive but not affirmative for DIC and it is recommended that it is repeated only occasionally.



**Presence of an underlying disorder
known to be associated with DIC
(no=0, yes=2)**



Global coagulation results

Platelet count ($> 100k = 0$, $< 100k = 1$, $< 50k = 2$)

Fibrin degradation products

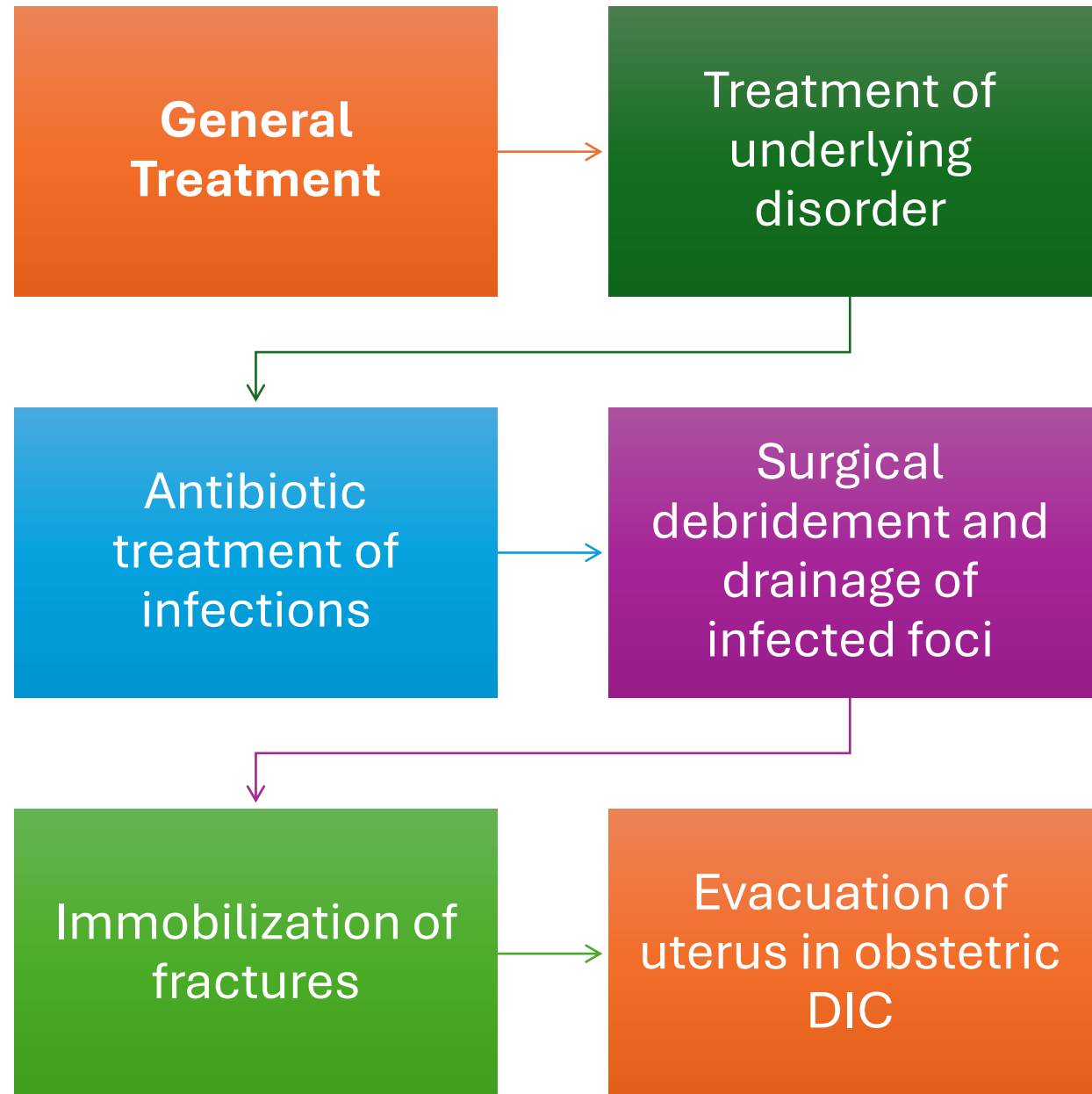
such as D-Dimer (no increase = 0,
moderate increase = 2, strong increase = 3)

Prolonged prothrombin time

($< 3 \text{ sec} = 0$, $> 3 \text{ sec} = 1$, $> 6 \text{ sec} = 2$)

Fibrinogen level ($> 1.0g/L = 0$; $< 1.0g/L = 1$)

Treatment



Supportive Treatment

Supportive treatment of MODS

Shock : fluids, catecholamines

Hypoxemia : oxygen, mechanical ventilation

Renal failure : diuretics, renal replacement therapy

Severe anemia : blood transfusion

Hemostatic Therapy

Antithrombotic treatment

AT concentrate.

Concurrent treatment with heparin should be avoided, heparin worsens thrombocytopenia

Fresh frozen plasma (FFP)

administer after

When bleeding;

antithrombin

Platelets : severe thrombocytopenia + bleeding

Antifibrinolytic treatment

Should be avoided

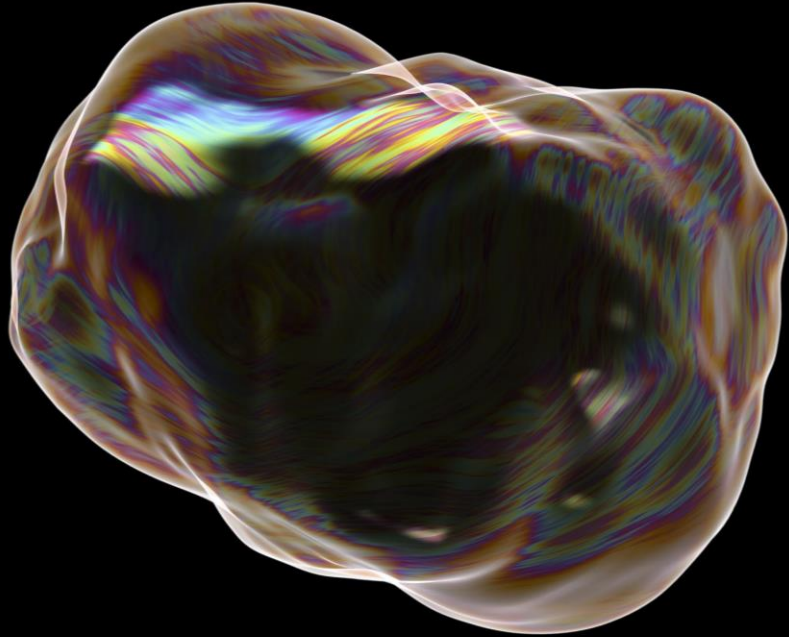
A quality antithrombin (AT)concentrate

Avoid heparin therapy with antithrombin concentration treatment

Nursing Care Plans & Management

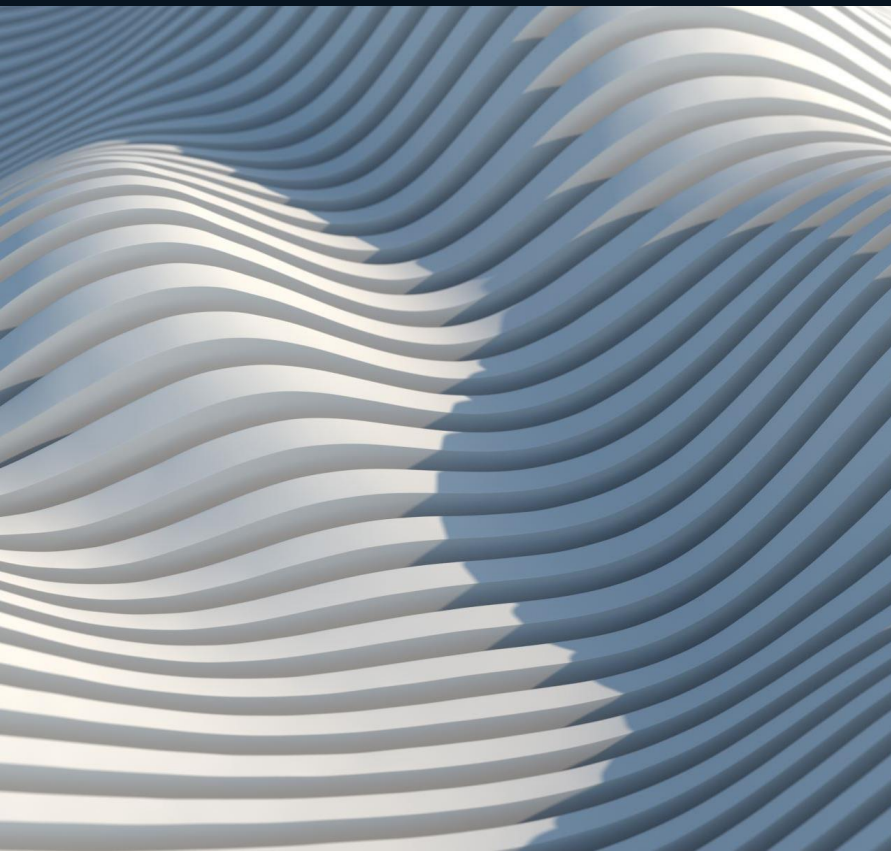


- **Nursing Assessment**
- ***Assess for the following subjective and objective data:***
- Abnormal arterial blood gases (ABGs)
- Abnormal breathing (rate, depth, and rhythm)
- Confusion
- Dyspnea
- Hypercapnia
- Hypoxemia
- Hypoxia
- Irritability
- Restlessness
- Somnolence
- Abnormal blood profile



- Capillary refill >3 seconds
- Changes in the level of consciousness
- Chest pain
- Cyanosis
- Hematuria
- Oliguria

***Assess for factors
related to the cause of
disseminated
intravascular
coagulation:***



The altered oxygen-carrying capacity of the blood

Blood circulation disruption

Microthrombi

Abnormal blood profile (depleted coagulation factors)

Drug therapy (adverse effects of heparin)

Complexity of treatment

Emotional state affecting learning

New condition and/or treatment

Unfamiliar environment

Nursing Diagnosis

The following are the nursing priorities for patients with disseminated intravascular coagulation DIC:

Minimizing bleeding risk

Respiratory support

Patient education



Nursing Interventions



Promoting Effective Gas Exchange



1. Assess for changes in the level of consciousness.

Early signs of cerebral hypoxia are restlessness and irritability; later signs are confusion and somnolence.



2. Assess the respiratory depth, rate, and rhythm.

The client will adapt breathing patterns over time to facilitate gas exchange.



3. Assess the client's breath sounds. Assess cough for signs of bloody sputum.

Changes in breath sounds may reveal the cause of impaired gas exchange. Hemoptysis is an indication of bleeding in the respiratory tract.

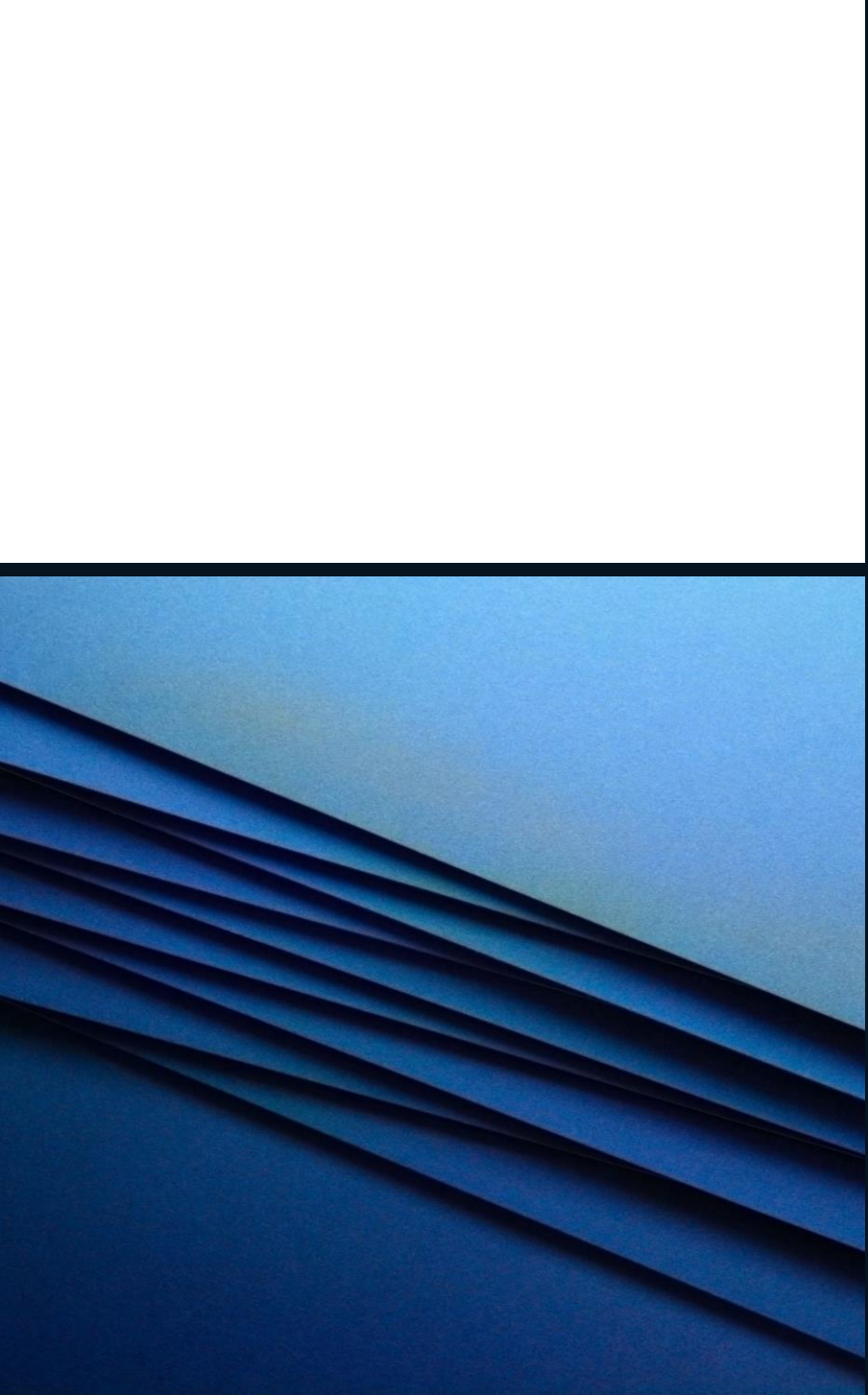
4. Assess for tachycardia, shortness of breath, and use of accessory muscles.

These signify an increased work of breathing. With initial hypoxia, HR increases.

5. Monitor oxygen saturation and assess arterial blood gases (ABGs).

Pulse oximetry is a useful tool to detect early changes in oxygen saturation.

Oxygen saturation should be kept at 90% or greater. Increasing PaCO₂ and decreasing PaO₂ are signs of hypoxemia and respiratory acidosis.



6. Provide reassurance and allay anxiety by staying with the client during the acute episodes of respiratory distress.

Anxiety increases dyspnea, the work of breathing, and the respiratory rate.

7. Change the client's positioning every 2 hours, and perform chest physiotherapy.

These maneuvers facilitate the movement and drainage of secretions.



- **8. Position the client in a high-Fowler's position as indicated.**

An upright position allows for adequate diaphragmatic and lung excursion and promotes optimal lung expansion.

- **9. Assist with coughing or suction as indicated**
- Productive coughing is the most effective way to remove moist secretions.
- If the client is unable to perform independently, suctioning may be needed to promote airway patency and reduce the work of breathing.

10. Maintain an oxygen administration device as ordered.

The appropriate amount of oxygen must be delivered continuously so that the client maintains an oxygen saturation of 90% or greater.

11. Anticipate the need for intubation and mechanical ventilation.

Early intubation and mechanical ventilation are recommended to prevent full decompensation of the client. Mechanical ventilation provides supportive care to maintain adequate oxygenation and ventilation for the client.

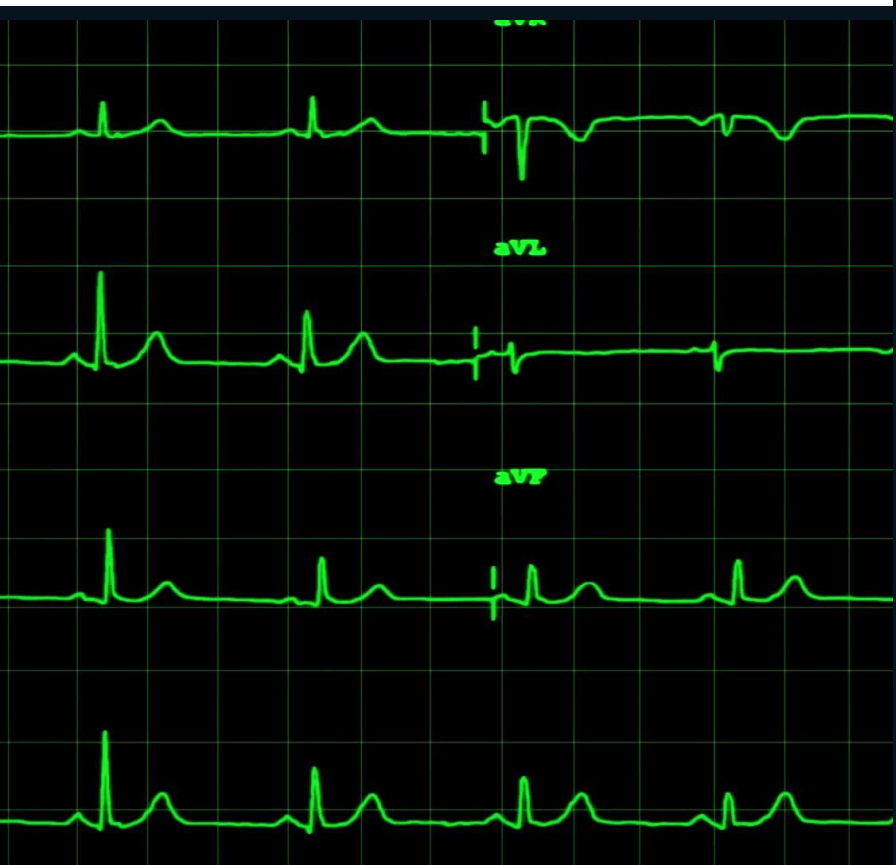


2. Enhancing Tissue Perfusion

- **1. Assess for contributing factors:**

When the proteins used in the normal clotting process become overly active, it can cause DIC.

- Infection, severe [trauma](#), inflammation, [surgery](#), Obstetrical complications (such as abruptio [placenta](#), and intrauterine fetal [death](#)), and [cancer](#) are all known to contribute to DIC.
-



- **2. Assess for the signs and symptoms of DIC.**
Bleeding, from mucous membranes, venipunctures sites, and areas from the gastrointestinal and [urinary](#) tracts.
- **3. Assess for [chest pain](#) and shortness of breath.**
Blood clots may form in the [blood vessels](#) of the [lungs](#) and [heart](#), therefore, blocking the blood flow that can cause these symptoms.
- **4. Assess the amount and color of [urine](#).**
Hematuria and [oliguria](#) occur ([urine](#) output less than 30 ml/hour) due to decreased perfusion to the [kidneys](#) as a result of tissue [injury](#) and clotted capillary beds.



5. Assess the client's level of consciousness.

A decreased level of consciousness can be precipitated by hemorrhagic changes or insufficient [oxygenation](#) of the [brain](#).

6. Assess arterial blood gases (ABGs).

Arterial blood gases may reveal a compensatory [respiratory alkalosis](#) in an attempt to decrease hydrogen ion concentration from hypoxia striking at the tissue level.

7. Monitor platelet count.

[Thrombocytopenia](#) (a common cause of abnormal bleeding, can occur as a result of insufficient platelet production by the [bone](#) marrow, or due to increased platelet destruction).

- 
- **8. Monitor PT (prothrombin time) and PTT (partial thromboplastin time).**

These laboratory values are typically prolonged as coagulation factors are consumed.

- **9. Monitor D-dimer levels.**

This is a test that detects a protein that results from clot breakdown; it is often markedly elevated with DIC.

- **10. Position the client in a semi-Fowler's to high-Fowler's as tolerated.**

Upright positioning promotes improved alveolar gas exchange.

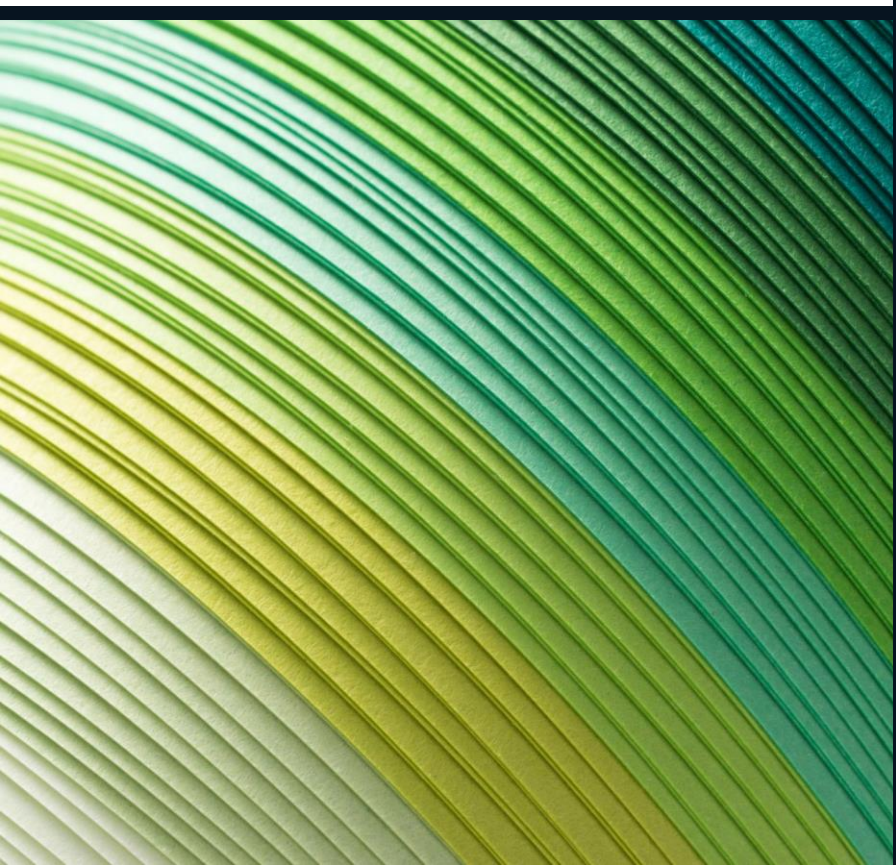


- **11. Provide oxygen therapy as necessary.**
This saturates circulating hemoglobin and augments the efficiency of blood that is reaching the ischemic tissues.
- **12. Administer parenteral fluids as prescribed.**
The maintenance of an adequate blood volume is vital for maintaining cardiac output and systemic perfusion.
- **13. Administer heparin as prescribed.**
Treatment with anticoagulant is used primarily to prevent the formation of new clots by decreasing the normal activity of the clotting mechanism.

3-Preventing Bleeding Risk and Injury



- **1. Assess for the underlying cause of DIC.** DIC is not a primary disease but occurs in response to a precipitating factor such as an infection or tumor.
- Successful treatment of DIC includes the management of the underlying disorder
- **Assess the client's heart rate and blood pressure.**
- **Observe for signs of orthostatic hypotension.** Tachycardia and hypotension are signs of decreased cardiac output.
- Orthostasis (a drop of more than 15 mm Hg when changing from a supine to a sitting position) indicates reduced circulating fluids.



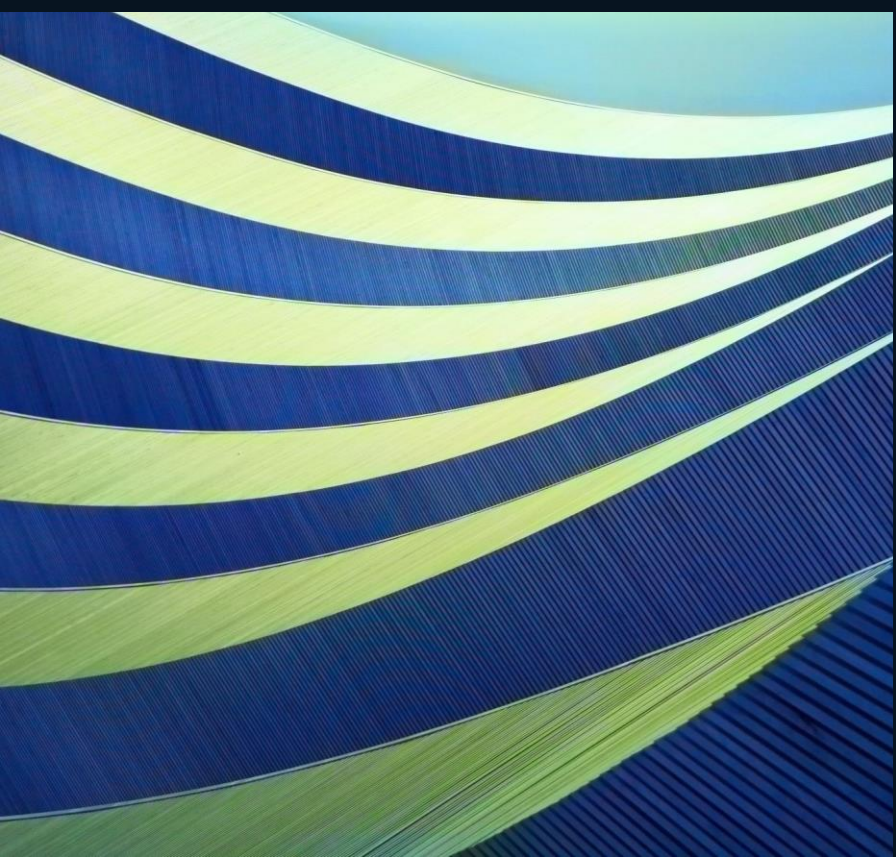
Observe for signs of internal bleeding, such as pain or changes in the level of consciousness. Institute a neurological checklist.

4. Observe for signs of external bleeding from the gastrointestinal (GI) and genitourinary (GU) tracts.

For example, the client may have increased skin bruising, hemoptysis, and hematuria.

5. Note any hemoptysis or blood obtained during suctioning.

These are the common manifestation of acute DIC.

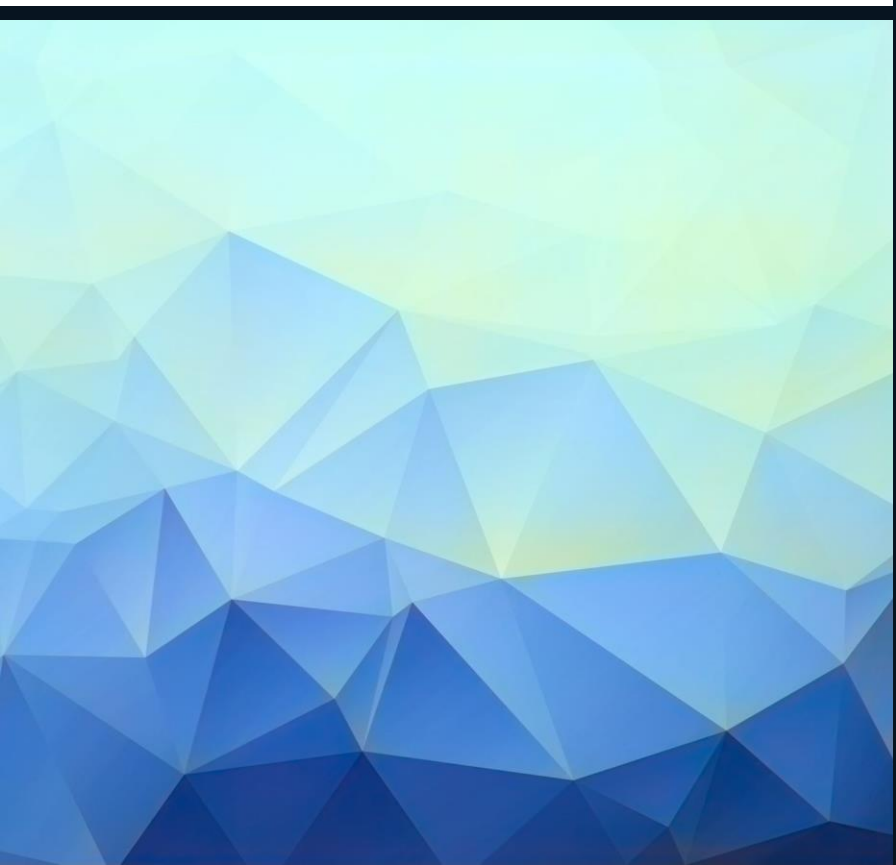


. Examine the skin surface for signs of bleeding. Note petechiae; purpura; hematomas; oozing of blood from IV sites, drains, and wounds; and bleeding from the mucous membranes.

Prolonged oozing of blood from injection sites or venipuncture sites could be the first indication of DIC.

7. Monitor hemoglobin and hematocrit levels.

Decreased hemoglobin and hematocrit levels are associated with bleeding from DIC.

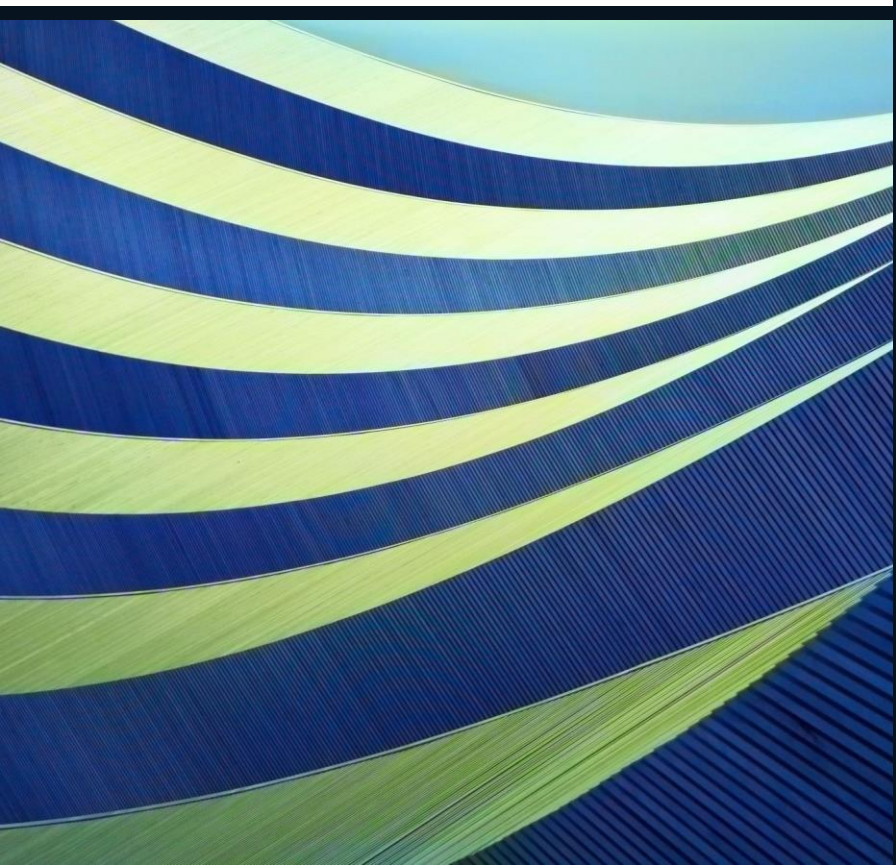


Monitor serial coagulation profiles.

Initially, accelerated clotting is noted.

Common laboratory values in DIC are PT greater than 15 seconds, PTT greater than 60 to 90 seconds, hypofibrinogenemia, thrombocytopenia, elevated fibrin split products (FSPs), elevated d-dimers, and prolonged bleeding time.

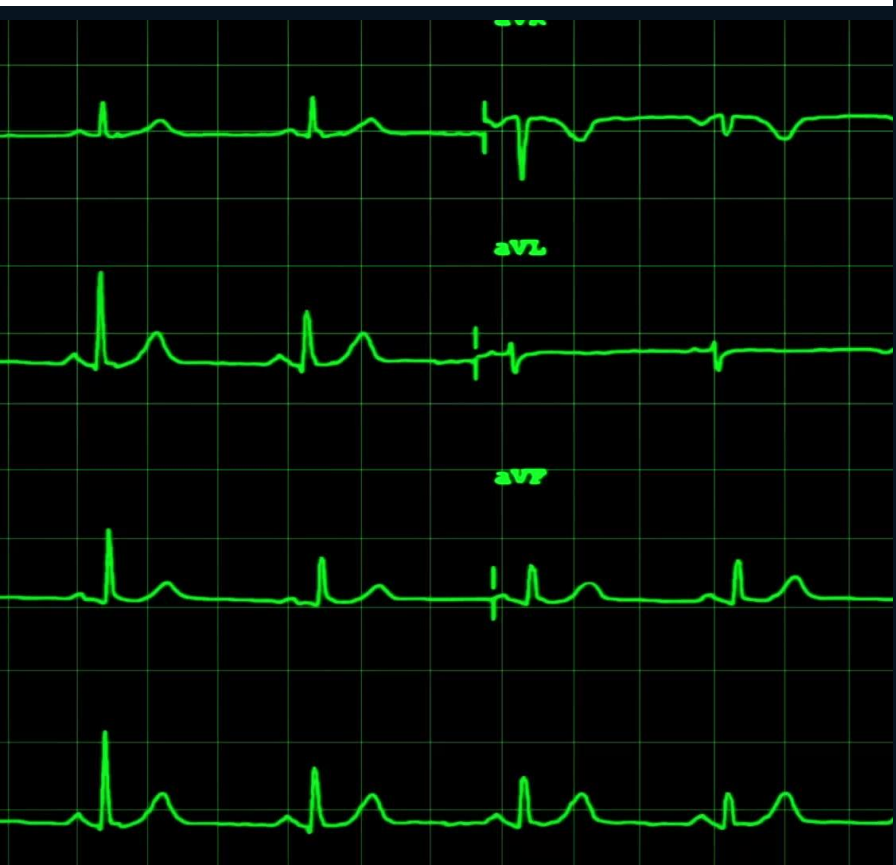
All put the client at risk for increased bleeding.



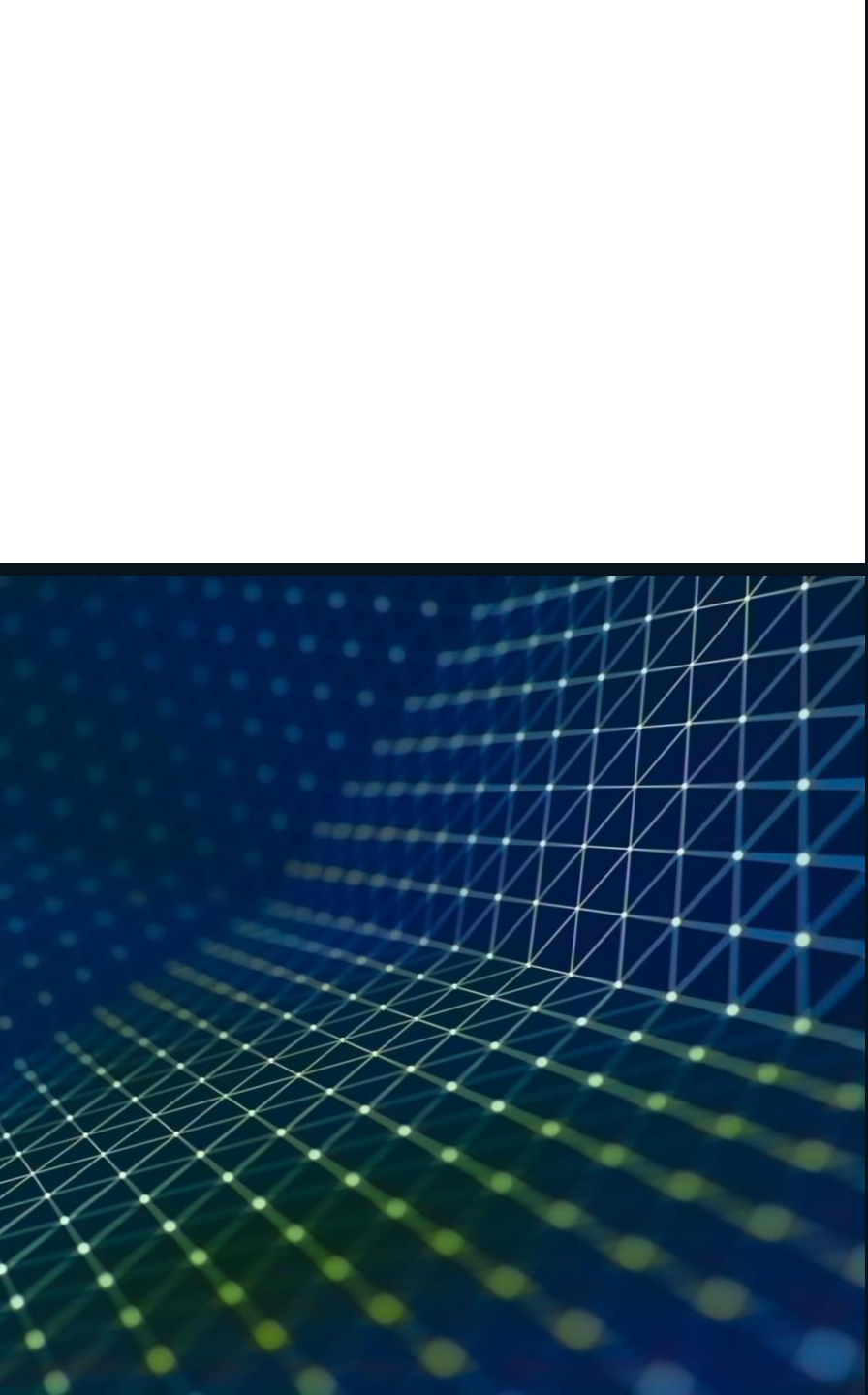
9. If heparin therapy is initiated, observe for an increase in bleeding from IV sites, GI/GU tracts, respiratory tract, or wounds; new purpura, petechiae, or hematoma.

Heparin is used for milder cases when clotting is more of a problem than bleeding.

It aborts the clotting process by blocking thrombin production.



- **10. Institute precautionary measures**
- such as avoiding intramuscular injections, avoiding unnecessary venipunctures
- Intramuscular injections can cause bleeding or hematoma formation, which is the accumulation of blood in the tissue.
- In DIC patients, this can lead to further clotting and organ damage.
- Therefore, must be avoided.
- Instead use alternative routes of medication administration, such as oral or subcutaneous routes, can be used in DIC patients.
- These routes are less invasive and put a lower risk of bleeding or hematoma formation.



11. Provide gentle oral care, using saline and water rinses instead of toothbrushes.

Maintaining good oral care is important in DIC patients to prevent oral infections, but some patients may not be able to tolerate toothbrushing due to the risk of bleeding.

In such cases, saline and water rinses can be used instead of toothbrushing.

12. Use gentle chest physiotherapies, such as turning, repositioning, coughing, deep breathing, percussion, and vibration.

Nursing interventions

- should be planned and implemented to eliminate potential sources of bleeding and to control the amount of potential bleeding and tissue injury.



13. Administer heparin therapy as prescribed.

If bleeding is increased, notify the physician of the possible need to decrease the IV drip.

Heparin is used for milder cases when clotting is more of a problem than bleeding.

Heparin augments antithrombin III activity that interrupts the clotting cycle and conversion of fibrinogen to fibrin.

It also blocks the intrinsic and extrinsic pathways by inhibiting factor X, which slows clot formation.

14. Administer parenteral fluids as prescribed.

for clients with hypotension.

The maintenance of an adequate blood volume is vital for maintaining cardiac output and systemic perfusion.

15. Administer blood products as prescribed: red blood cells (RBCs), fresh frozen plasma (FFP), and platelets.

Blood and plasma transfusions replace blood clotting factors.

RBCs increase oxygen-carrying capacity; FFP replaces clotting factors and inhibitors; platelet and cryoprecipitate provide proteins for coagulation.

16. Administer additional medications or investigational drugs as ordered.



4. Initiating Health Teachings and Patient Education

- 1. Assess the client's knowledge of DIC.**
- 2. Carefully explain the underlying cause that precipitated DIC.**
- 3. Explain the purpose of drug and transfusion therapy.**
The controversial nature of treatment may be difficult for the client or significant others to understand in the acute setting.

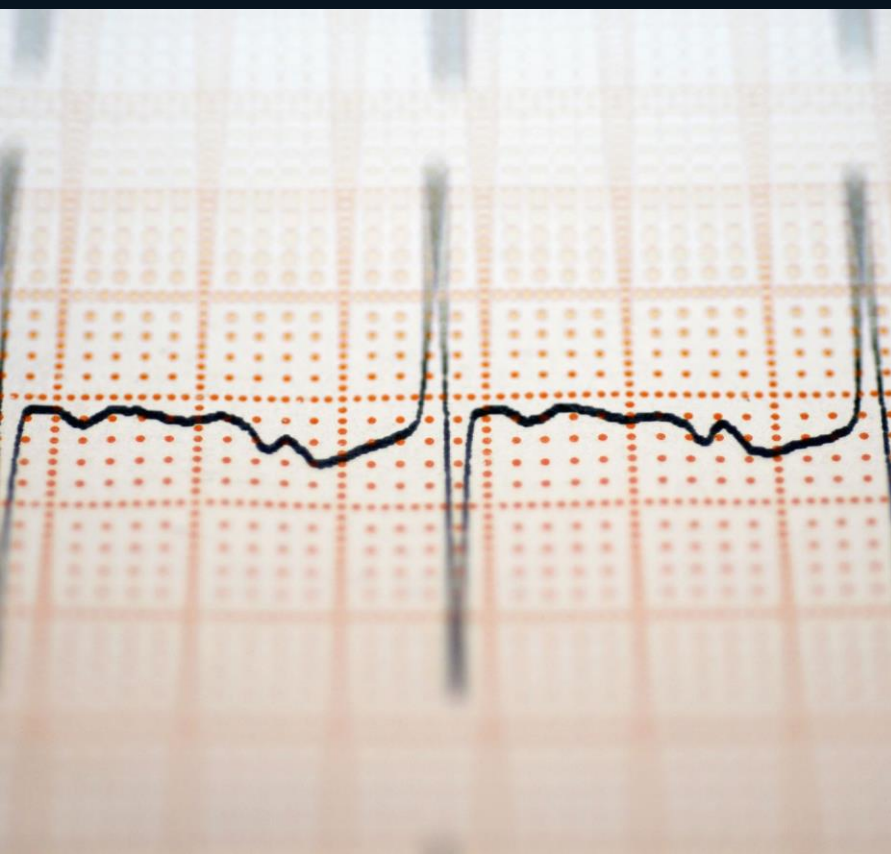
4. Instruct the client or significant others to notify the nurse of new bleeding from wounds or IV sites.

This notification can aid in achieving early intervention at bleeding sites.

5. Assessing and Monitoring for Potential Complications

1. Monitor patient's vital signs regularly.

Regularly assessing vital signs such as blood pressure, heart rate, respiratory rate, and temperature helps identify any abnormalities that may indicate the presence of complications



- **2. Monitor patient's [cardiac rhythm](#).**
Continuous monitoring of the patient's cardiac rhythm and electrocardiogram ([ECG](#)) helps identify any arrhythmias or changes in cardiac function.
- **3. Assess patient's neurological status.**
Regular assessment of neurological status, including level of consciousness, motor strength, and sensory function, helps identify any changes that may indicate compromised perfusion to the [brain](#).
- **4. Assess patient's respiratory status.**
Monitoring respiratory rate, [oxygen saturation](#), and lung sounds helps identify any signs of respiratory distress or compromised [gas exchange](#).

8. Assess the patient's medication regimen and monitor for potential side effects or interactions.

Some medications used in the management of DIC, such as [anticoagulants](#) or blood products, carry inherent risks and require close monitoring.

9. Collaborate with interdisciplinary team.

Collaborating with other healthcare professionals, such as physicians, pharmacists, and laboratory staff, ensures comprehensive assessment and monitoring.

- **10. Provide information about the disease to patient and family members.**

Providing education to patients and their families about potential complications, signs and symptoms to watch for, and the importance of reporting any changes promptly is essential.

- Educating patients on self-monitoring and encouraging compliance with follow-up appointments promotes their active involvement in their care.

5. Recombinant human-activated protein C

This inhibits factors Va and VIIIa of the coagulation cascade.

6. Platelet transfusions

Platelet transfusions are used to increase the platelet count and improve clotting in patients with severe thrombocytopenia in DIC.

7. Fresh frozen plasma (FFP)

FFP is a blood product that contains various clotting factors.



- **7. Monitoring Laboratory and Diagnostic Procedures**
- **1. Complete Blood Count (CBC)**
- **2. Coagulation Profiles**
Coagulation profiles, such as prothrombin time (PT), activated partial thromboplastin time (aPTT), and international normalized ratio (INR),
- help evaluate the patient's clotting ability.
Abnormal results indicate coagulation abnormalities associated with DIC.

- **3. Fibrinogen Level**

Fibrinogen is a key clotting factor.

- Monitoring fibrinogen levels helps assess the patient's clotting capacity and indicates the progression or resolution of DIC.

- **4. D-dimer Assay**

D-dimer is a degradation product of fibrin clots.

- Elevated D-dimer levels suggest ongoing fibrinolysis and clot breakdown, which are commonly observed in DIC.

- **5. Liver and Renal Function Tests**

DIC can affect liver and kidney function. Monitoring liver enzymes (such as ALT,

- 
- AST) and [renal](#) function (such as serum [creatinine](#)) helps assess organ involvement and guide management decisions.

- **6. Imaging Studies**

Imaging modalities such as ultrasound, [computed tomography](#) (CT) scans, or [magnetic resonance imaging](#) (MRI) may be used to evaluate organ damage or identify [thrombus](#) formation in DIC.

- **7. [Arterial Blood Gas Analysis](#)**

[Arterial blood gas](#) analysis helps assess the patient's [acid-base balance](#) and [oxygenation](#) status, providing important information about respiratory function and tissue perfusion.

Post test



Mention causes of DIC



Mention diagnostic studies to
assess coagulation profile



Any question
Thank you